

Sequences

Answer sheet · 26 marks total

Section A — Term-to-term rules

A1. **32** (1)

A2. **36** (1)

A3. **Multiply by 2** (1)

A4. **Subtract 9** (1)

Section B — Special sequences

B1. **1, 4, 9, 16, 25** (2, 1 if one error)

B2. **1, 3, 6, 10, 15** (2, 1 if one error)

B3. **21, 34** (1 each)

Section C — Finding the n th term

C1. $d = 3$ (1) $a - d = 2 - 3 = -1$ (1) $= 3n - 1$ (1)

C2. $d = -7$ (1) $a - d = 40 - (-7) = 47$ (1) $= -7n + 47$ (1)

Section D — Using the n th term

D1. $4(12) + 5$ (1) $= 53$ (1)

D2. $6n - 4 = 100 \rightarrow 6n = 104 \rightarrow n = 17.33\ldots$ (1) not a whole number (1) so **No** (1)

D3. $d = 5$ (1) $a - d = 9 - 5 = 4$ (1) n th term $= 5n + 4$ (1) 20th term: $5(20) + 4$ (1) $= 104$ (1)

Marking note: award method marks for correctly identifying the common difference or correctly substituting into a formula, even if the final answer contains an arithmetic error.